

AMAZON

Sales Analysis

Amazon Products | Power BI Data Analytics Project

Project Focus	Sales performance, product performance, time-based trends and interactive KPI reporting.
Dashboard	Power BI dashboard with Total Sales, QTD Sales, YTD Products Sold, YTD Reviews, monthly and weekly trends, category analysis and Top 5 products.
Data Model	Amazon_Data fact table connected to a dedicated Date Table for time intelligence.

Prepared as a recruiter-friendly project report

The structure follows the user's earlier Customer Shopping Behavior Analysis report while using the new Amazon Sales Analysis workflow and dashboard.

1. Project Overview

This project analyzes Amazon product sales data and converts the raw transaction data into an interactive Power BI dashboard. The objective is to understand overall sales performance, quarterly and year-to-date metrics, product/category contribution, and time-based sales trends.

2. Dataset & Business Scope

Dataset / Main Table	Amazon_Data
Key fields visible in the model	Order Date, Price(Dollar), Product Category, Product Description, Shipment, Number of reviews
Time period represented	2019–2022, based on the Date Table year values
Business questions	How much was sold? Which months/weeks perform strongly? Which categories and products contribute most? How do results change with year/category/quarter filters?

3. Project Workflow

- **Data preparation:** Load the Amazon sales dataset and verify the available fields and data types.
- **Data modeling:** Build a dedicated Date Table and establish the relationship between Date Table[Date] and Amazon_Data[Order Date].
- **Time intelligence:** Create Month, Month Number, Quarter, Week Number and Year fields.
- **DAX measures:** Create Total Sales, QTD Sales, YTD Products Sold, YTD Reviews and related analytical measures.
- **Dashboard design:** Combine KPI cards, slicers, a monthly trend, weekly sales chart, category table and Top 5 product charts.

4. Data Preparation & Modeling

The Power BI model was organized around a transaction table and a separate calendar-style Date Table. This structure supports clean filtering and time-based calculations.

4.1 Date Table

Date Table was created from the minimum and maximum Order Date values:

```
Date Table = CALENDAR(MIN('Amazon_Data'[Order Date]), MAX('Amazon_Data'[Order Date]))
```

4.2 New Date Fields

Field	DAX / Logic	Purpose
Month	FORMAT(Date, "MMM")	Displays Jan, Feb, Mar ... Dec
Month Number	MONTH(Date)	Sorts months chronologically
Qtr	CONCATENATE("Qtr-", Quarter Number)	Displays Qtr-1 ... Qtr-4
Quarter Number	QUARTER(Date)	Returns quarter number 1–4
Week Number	WEEKNUM(Date)	Groups sales by week
Year	YEAR(Date)	Provides 2019–2022 year filter

Important visual setting: Month is sorted by Month Number so that charts display Jan → Dec instead of alphabetical order.

5. DAX Measures & KPI Logic

The dashboard uses measures to calculate business KPIs dynamically. The Total Sales measure is used for the overall revenue card, while QTD and YTD measures support time-based reporting.

Measure	DAX / Logic	Dashboard use
Total Sales	SUM(Amazon_Data[Price(Dollar)])	Overall sales; responds to selected filters
YTD Sales	TOTALYTD(SUM(Amazon_Data[Price(Dollar)]), 'Date Table'[Date])	Year-to-date sales analysis
QTD Sales	Quarter-to-date sales measure	Quarterly sales KPI
YTD Products Sold	Year-to-date product quantity measure	Products sold KPI
YTD Reviews	Year-to-date reviews measure	Review-volume KPI

5.1 Why Total Sales and YTD Sales are separate

Total Sales returns the sum of sales in the current filter context. YTD Sales is a time-intelligence calculation that evaluates sales year-to-date. Keeping these as separate measures prevents a multi-year selection from being mistaken for a single year's YTD value.

TOTAL SALES	\$8.42M	QTD SALES	\$811.09K
Dashboard snapshot		Dashboard snapshot	
YTD PRODUCTS SOLD	27.75K	YTD REVIEWS	19.42M
Dashboard snapshot		Dashboard snapshot	

6. Dashboard Requirements & Visual Design

The dashboard was designed to answer the main business questions in a single interactive report page.

Visual	Purpose
Total Sales KPI	Monitor overall sales across the selected filter context.
QTD Sales KPI	Monitor quarter-to-date sales performance.
YTD Products Sold	Track product volume on a year-to-date basis.
YTD Reviews	Track review volume on a year-to-date basis.
Sales by Month	Identify monthly seasonality and changes in sales volume.
Sales by Week	Identify shorter-term weekly fluctuations.
Sales by Sub Category	Compare category-level YTD sales, QTD sales and percentage contribution.
Top 5 Products by YTD Sales	Highlight the leading products by sales.
Top 5 Products by YTD Reviews	Highlight products receiving the highest review volume.

Interactive Filters

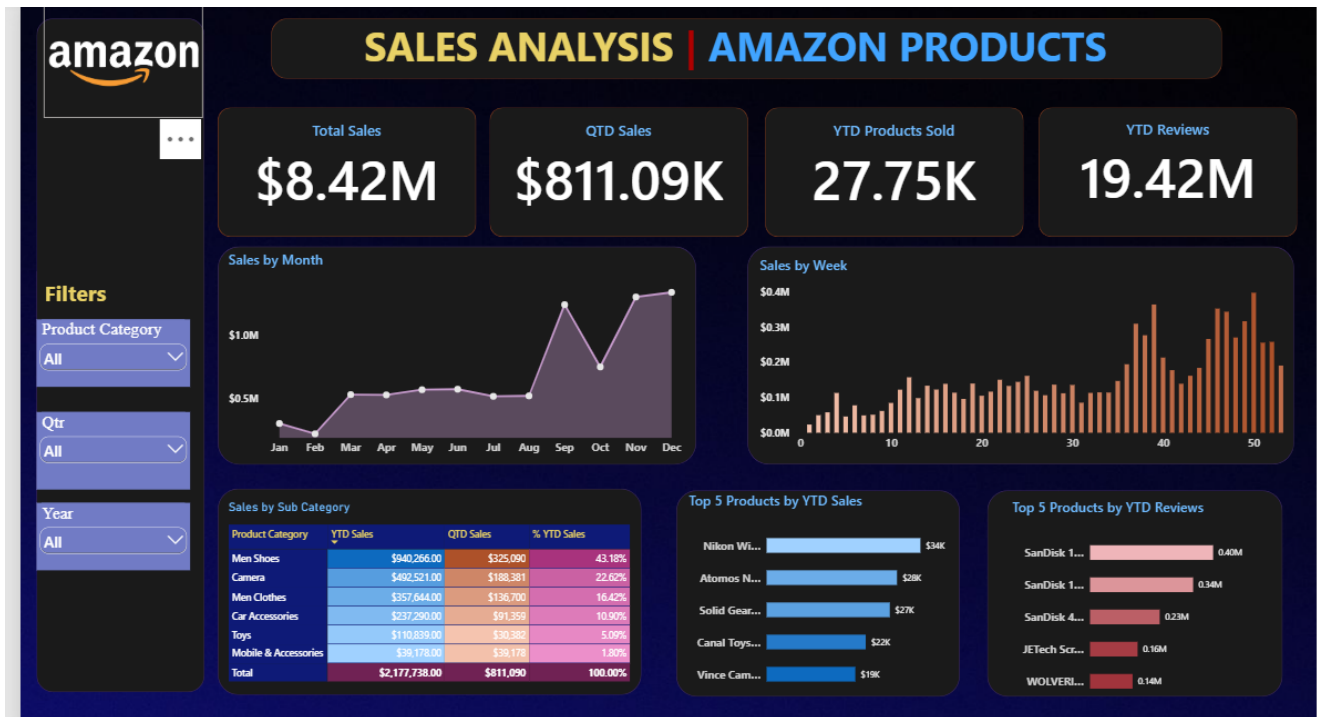
- **Product Category:** filters the dashboard by product category.
- **Qtr:** filters the report by quarter.
- **Year:** filters the report by year; multiple years can be selected.

Dashboard design principles

- KPI cards are placed at the top for immediate visibility.
- Trend charts occupy the middle area to show time-based movement.
- Category and Top 5 product visuals provide focused product-level analysis.
- The dark dashboard screenshot is retained as the actual project output; this report uses a light premium presentation theme.

7. Dashboard Snapshot

The following image is the actual dashboard supplied for the Amazon Sales Analysis project. It is included as a full-page visual so the report documents the final Power BI output.



Dashboard snapshot shown with Total Sales, QTD Sales, YTD Products Sold, YTD Reviews, Sales by Month, Sales by Week, Sales by Sub Category, Top 5 Products by YTD Sales and Top 5 Products by YTD Reviews.

8. Key Insights from the Dashboard

The following observations are directly visible in the supplied dashboard snapshot.

Overall sales	Total Sales is shown as \$8.42M in the dashboard snapshot.
Quarter performance	QTD Sales is shown as \$811.09K in the dashboard snapshot.
Monthly trend	Sales show a noticeable increase in September and again in November–December, while February is comparatively lower in the displayed annual trend.
Category contribution	In the displayed category table, Men Shoes has the largest YTD Sales contribution at 43.18% .
Top product sales	The Top 5 Products by YTD Sales visual shows the leading product at approximately \$34K , followed by products around \$28K, \$27K, \$22K and \$19K.
Review volume	The Top 5 Products by YTD Reviews visual shows the leading product at approximately 0.40M reviews.

9. Business Recommendations

Recommendations below are framed from the patterns visible in the dashboard and are intended as business actions to investigate further.

1. Prioritize high-contribution categories

Use category-level sales contribution to focus merchandising, inventory planning and promotional activity on categories generating the largest share of sales.

2. Plan around seasonal peaks

The monthly trend indicates stronger sales around September and November–December. Marketing and inventory planning can be aligned with these periods.

3. Study Top 5 product drivers

Review the products appearing in the Top 5 YTD Sales visual to understand what characteristics, pricing or promotions may be associated with higher sales.

4. Use review volume as a customer-signal metric

Monitor high-review products alongside sales performance to identify products with strong customer engagement and opportunities for further promotion.

5. Use slicers for management questions

Apply Product Category, Quarter and Year filters to compare performance across different business periods and segments.

10. Conclusion

The Amazon Sales Analysis project demonstrates an end-to-end Power BI reporting workflow: organizing sales data, creating a dedicated Date Table, building time-intelligence fields and measures, and presenting the results through an interactive dashboard. The final report makes the model, KPI logic, visual requirements and business observations easy to communicate to recruiters, instructors and business stakeholders.

Project Deliverable
Amazon Sales Analysis – Power BI Dashboard & Project Report